

Project Demonstration

Credit Card Approval Prediction System | Project Demonstration Phase

Demonstration Objective

Showcase the end-to-end Credit Card Approval Prediction System: from filling the 17-field applicant form to receiving an Approved or Rejected decision on the result page.

How to Run Locally

Step	Command / Action
1	Install dependencies: <code>pip install -r requirements.txt</code>
2	Train the model (first time only): <code>python train_model.py</code>
3	Start the Flask app: <code>python app.py</code>
4	Open browser: <code>http://127.0.0.1:5000/</code>
5	Fill all 17 fields and click 'Analyze Application'
6	View result on result.html — green (Approved) or red (Rejected)
7	Click '← Analyze Another Applicant' to return to form
Prod	Deploy on Render: <code>gunicorn app:app</code> (see render.yaml)

Sample Walkthroughs

Scenario	Key Features	Expected Output
Low-Risk Applicant (Approved)	Female, 35 yrs, married, owns car+realty, income=150,000, High Credit Quality	Approved (green card)
High-Risk Applicant (Rejected)	Male, 52 yrs, single, no car, income=25,000, Secondary Education, 10 yrs employed, STATUS his	Rejected (red card)
Unemployed Applicant	Experience = 0 years → DAYS_EMPLOYED = 365243 (sentinel value)	Prediction based on other features; no error
Unknown Occupation	OCCUPATION_TYPE = Unknown	OHE handle_unknown='ignore' → prediction s

What the Demonstration Covers

- Dark-theme form UI with two sections: Personal Profile and Financial & Professional Profile.
- Friendly input mode (age in years + experience in years) with automatic JS conversion.
- Toggle to Raw Days Mode for technical users.
- Approved result — green tick card with 'Low credit risk detected.' message.
- Rejected result — red cross card with 'High risk variance detected.' message.
- Error case — blank required field triggers HTML5 validation message.

- Live Render deployment — public URL accessible from any browser.

Demonstration Artifacts

- Screen recording of the complete form-to-result walkthrough (attach separately).
- Screenshots of Approved and Rejected result pages.
- Screenshot of the friendly-to-raw toggle mode.
- Live Render URL shared with the mentor/evaluator.